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Assignment - 2

Date

Q1) (i) If $n=5$, $p=0.20$ then what will be the mean and variance of Binomial distribution
 $n=5$, $p=0.20$

Mean $\rightarrow \mu = n \cdot p = 5 \times 0.20 = 1$

Variance $\rightarrow \sigma^2 = npq = np(1-p)$
 $= 5 \times 0.2(1-0.2)$
 $= 0.80$

(ii) The mean and variance of Binomial distribution are 4 and $4/3$ respectively. find $P(X=1)$
 (Ans = 0.99863)

$\mu = 4$ $\sigma^2 = 4/3$

$4 = np$

$p = 4/n$

$4/3 = np(1-p)$

$4/3 = np(1-p)$

$4/3 = n \times \frac{4}{n} \left(1 - \frac{4}{n}\right)$

$\frac{1}{3} = 1 - \frac{4}{n}$

$\frac{4}{n} = 2/3$

$n = 6$

$\therefore p = 4/6 = 2/3$

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$P(X \geq 1) = 1 - P(X=0)$

$= 1 - {}^6C_0 p^0 (1-p)^6$

$= 1 - 1 \times 1 (1-2/3)^6$

$= 1 - (1/3)^6$

$= 1 - 1/729 = \frac{728}{729} = 0.99863$

(iii) Comment on the following statement: "The mean of a Binomial distribution is 5 and the standard deviation is 3"

$\mu = np = 5$

$\sigma^2 = np(1-p)$

$\sigma = \sqrt{np(1-p)}$

$3 = \sqrt{np(1-p)}$

$9 = 5(1-p)$

$\frac{9}{5} = 1-p$

$-9/5 = -p$

Statement can't be true for a Binomial distribution because it leads to invalid probability

(iv) If X is a Poisson variate such that $P(X=2) = 9P(X=4) + 90P(X=6)$. find mean

variance of X

As $\frac{\lambda^2 e^{-\lambda}}{2!} = 9 \frac{\lambda^4 e^{-\lambda}}{4!} + 90 \frac{\lambda^6 e^{-\lambda}}{6!}$

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$$3) \frac{d^2}{2} = \frac{9x^4}{24} + \frac{50 \cdot 2^6}{720}$$

$$3) \frac{d^2}{2} = \frac{d^2}{2} \left(\frac{9 \cdot 3^4}{4} + \frac{d^4}{84} \right)$$

$$\rightarrow 4 = 3d^2 + d^4$$

$$\rightarrow d^4 + 3d^2 - 4 = 0$$

$$\rightarrow y^2 + 3y - 4 = 0$$

$$\rightarrow y^2 + 4y - y - 4 = 0$$

$$\rightarrow y(y+4) - 1(y+4)$$

$$\rightarrow (y+4)(y-1)$$

$$\rightarrow y = 1, -4$$

$$\therefore d^2 = -4 \text{ or } d^2 = 1$$

∴ mean = variance = positive distribution = 1

(i) If $n=4$ and $p=0.25$ then the mean and variance of binomial distribution will be...

$$n=4, p=0.25$$

$$\lambda = np = 4 \times 0.25 = 1$$

(ii) Under normal curve total area right side of mean is

The curve is symmetric about the mean. The total area under the curve represents the prob. of all possible outcomes, which sums to 1 (or 100%). ∴ area right side of mean is (i.e. $z=0$) is 50% of the total area.

Date

(vii) Mean and variance of standard normal
variate are --
Mean = 0
variance = 1

(viii) what is the pdf for uniform distribution
 $f(x) = \begin{cases} \frac{1}{b-a} & \text{for } 0 \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$

$$\text{Mean} = \frac{a+b}{2} \quad \text{variance} = \frac{(b-a)^2}{12}$$

(ix) The mean and variance of uniform distribution are 2 and $4/3$ respectively. Find $P(x < 0)$.

$$\mu = 2 = \frac{a+b}{2} \Rightarrow a+b = 4 \quad \text{--- (1)}$$

$$\sigma^2 = 4/3 = \frac{(b-a)^2}{12} \Rightarrow (b-a)^2 = 16$$

$$= (b-a) = 4 \quad \text{--- (2)}$$

∴ from (1) and (2)

$$(1) - (2)$$

$$a+b + b-a = 4 \Rightarrow b = 3$$

$$\therefore a = -1$$

$$\therefore \text{interval} = [a, b] = [-1, 3]$$

$$f(x) = \frac{1}{b-a} = \frac{1}{3-(-1)} = 1/4$$

$$P(x < 0) = \int_{-1}^0 f(x) dx = \int_{-1}^0 1/4 dx$$

$$P(x < 0) = 1/4 \times (0 - (-1)) = 1/4 \times 1 = 1/4$$

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(ix) what is the pdf of exponential distribution

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\text{Mean} = 1/\lambda \quad \text{variance} = 1/\lambda^2$$

(xi) In which condition mean and variance of exponential distribution are equal.

$$\lambda = 1$$

$$\sigma^2 = 1/\lambda^2$$

$$\text{if } \mu = \sigma^2 \Rightarrow \frac{1}{\lambda} = \frac{1}{\lambda^2}$$

$$\lambda^2 = 1$$

$$\lambda = 1$$

$$\lambda = 1$$

$$\lambda = 1, \lambda = 1$$

$\rightarrow \lambda$ must be > 0 $\because$ it represents rate of success

$\rightarrow \lambda = 1$ is valid

In this case $\mu = 1, \sigma^2 = 1$

(xii) which continuous distribution follows memoryless property?

Exponential distribution is the only continuous distribution that has memoryless property

Date

(Q2) Out of 800 families with 4 children each, how many families would be expected to have

(i) 2 boys and 2 girls

(ii) at least one boy

(iii) no girl

(iv) at most two girls

birth of children = independent events

$$\begin{aligned} P(X=2) &= {}^4C_2 (0.5)^2 (0.5)^{4-2} \\ &= \frac{4!}{2!(4-2)!} \times (0.5)^2 \times (0.5)^2 \\ &= 6 \times 0.25 \times 0.25 = 0.375 \end{aligned}$$

$$\text{for 800 families} = 0.375 \times 800 = 300$$

$$\begin{aligned} P(\text{at least one boy}) &= 1 - P(X=0) \\ &= 1 - {}^4C_0 (0.5)^0 (0.5)^4 \\ &= 1 - 0.0625 \\ &= 0.9375 \end{aligned}$$

$$\text{for 800 families} \Rightarrow 0.9375 \times 800 = 750$$

$$\begin{aligned} P(X=4) &= {}^4C_4 (0.5)^4 = 1 \times 0.0625 = 0.0625 \\ &\Rightarrow 0.0625 \times 800 = 50 \end{aligned}$$

$$P(X=4) + P(X=3) + P(X=2)$$

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$$\rightarrow 0.0025 + 4 \left(\frac{1}{2} (0.5)^3 (0.5)^1 \right) + 0.375$$

$$\Rightarrow 0.6975$$

$$\Rightarrow 0.6975 \times 800 = 558$$

Q3) 20% screws produced in a certain factory turn out to be defective. Find the prob. that in a sample of 15 screws chosen at random (exactly) four will be defective.

$$P(n=4) = {}^n C_k p^k (1-p)^{n-k}$$

$$n=15$$

$$k=4 \quad P(n=4) = {}^{15} C_4 (0.2)^4 \times (0.8)^{15-4}$$

$$p=0.2$$

$$q=1-p$$

$$= 1-0.2$$

$$= 0.8$$

$$= {}^{1365} \times (0.0016) \times (0.8)^{11}$$

$$= 0.1876$$

$\therefore$ 18.76% chance of 4 out of 15 randomly chosen screws

Q4) The prob. that a bomb dropped from a plane will strike the target is $1/5$. If some bombs are dropped, find the prob. that at least two will strike the target and exactly two will strike the target.

$$p=0.2 \quad q=1-p=1-0.2=0.8 \quad n=6$$

$$P(n=4) = {}^n C_k p^k (1-p)^{n-k}$$

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Q1) Prob. of atleast two bombs striking the target

$$P(n \geq 2) = 1 - P(n=0) + P(n=1)$$

$$P(n=0) = {}^6 C_0 (0.2)^0 (0.8)^6 = 1 \times 1 \times (0.8)^6 = 0.262144$$

$$P(n=1) = {}^6 C_1 (0.2)^1 (0.8)^5 = 6 \times 0.2 \times (0.8)^5 = 0.393216$$

$$P(n \geq 2) = 1 - [0.262144 + 0.393216] = 1 - 0.65536 = 0.34464$$

Q2) Prob. of exactly two bombs striking the target

$$P(n=2) = {}^6 C_2 (0.2)^2 (0.8)^{6-2} = 15 \times (0.2)^2 \times (0.8)^4 = 0.24576$$

Q5) In a book of 520 pages, 350 type errors occur. Assuming Poisson law for the number of errors per page, find the prob. that a random sample of 5 pages will contain no error.

$$\lambda_{\text{page}} = \frac{\text{total errors}}{\text{total page}} = \frac{350}{520} = 0.75$$

$$\lambda_s = 0.75 \times 5 = 3.75$$

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$$P(n=0) = \frac{(3.75)^0 e^{-3.75}}{0!} = e^{-3.75}$$

prob of random 5 pages containing no error

- Q6) Six coins are tossed 8400 times. Using the poisson distribution, find the approximate prob of getting 6 heads 2 times

$$P(n=k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \begin{matrix} \lambda = 6 \\ k = 2 \end{matrix}$$

$$P(6 \text{ heads in one}) = \frac{(1)^6}{2!} = 1/64$$

prob of 6 heads

$$\lambda = 8400 \times \frac{1}{64} = 100 \text{ is expected no.}$$

of times we will get 6 heads in 8400 tosses

$$P(n=2) = 100^2 e^{-100} / 2!$$

$$= 10000 \times 1 - 100 / 2 = e^{-100} \times 5000$$

- Q7) A box contains 100 cellophane, 20 of which are defective. Ten cellophane have been selected for inspection. find the prob that
- at least one is defective
 - all are defective
 - at the most 3 are defective
 - none is defective

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$$P(n=k) = {}^n C_k p^k (1-p)^{n-k}$$

$$n = 10$$

$$p = 0.2$$

$$1-p = 0.8$$

- (i) atleast one is defective

$$P(n=0) = {}^{10} C_0 (0.2)^0 (0.8)^{10} = 1 \times 1 \times 0.8^{10} = 0.1074$$

$$P(n \geq 1) = 1 - P(n=0) = 1 - 0.1074 = 0.8926$$

- (ii) Prob that at most 3 are defective

$$\begin{aligned} P(n \leq 3) &= P(n=0) + P(n=1) + P(n=2) + P(n=3) \\ &= {}^{10} C_0 (0.2)^0 (0.8)^{10} + {}^{10} C_1 (0.2)^1 (0.8)^9 + {}^{10} C_2 (0.2)^2 (0.8)^8 + {}^{10} C_3 (0.2)^3 (0.8)^7 \\ &= 0.1074 + 0.2684 + 0.3440 + 0.2013 \\ &= 0.9211 \end{aligned}$$

- (iii) Prob that none are defective

$$P(n=0) = {}^{10} C_0 (0.2)^0 (0.8)^{10} = 0.1074$$

- (iv) Prob that all are defective

$$\begin{aligned} P(n=10) &= {}^{10} C_{10} (0.2)^{10} (0.8)^0 \\ &= 1 \times (0.2)^{10} = 0.0000001024 \end{aligned}$$

Date

Q12) In a certain examination 2000 students appeared. The average marks obtained were 50% and standard deviation was 5%. How many students do you expect to obtain more than 60 marks, assuming the marks to be distributed normally.

$$\mu = 50 \quad P(X > 60) = ?$$

$$\sigma = 5$$

$$Z = \frac{X - \mu}{\sigma} = \frac{60 - 50}{5} = 2$$

$$\text{from table } P(Z \leq 2) = 0.9772$$

$$P(X > 60) = 1 - 0.9772 = 0.0228$$

$$\text{Expected no. of students} = 0.0228 \times 2000$$

$$= 45.6 \approx 46$$

Q13) Subway trains on a certain line run every half hour from midnight and on in the morning. What is the prob. that a person entering the station at random time during this period will have to wait at least twenty minutes?

total time span = 12 to 6 = 300 min
train in 30 min i.e. 12 trains
man will have to wait atleast 20 min if his arrival falls within (0, 20] min after train departs. for interval = $\frac{30-20}{30} = \frac{10}{30} = \frac{1}{3}$

$$P(X \geq 20) = \frac{1}{3}$$
Date

Q14) A random sample of size 100 is taken from an inf. population having mean 75 and variance 256. what is the prob. that sample mean will be b/w 75 and 78?

$$\mu = 75 \quad n = 100$$

$$\sigma^2 = 256 \Rightarrow \sigma = \sqrt{256} = 16$$

$$P(75 \leq \bar{X} \leq 78) = ?$$

$$\text{for } 75, Z_1 = \frac{75 - 75}{16/\sqrt{100}} = \frac{-1}{16 \times 10} = -0.025$$

$$\text{for } 78, Z_2 = \frac{78 - 75}{16/\sqrt{100}} = \frac{3}{16 \times 10} = 1.25$$

$$P(75 \leq \bar{X} \leq 78) = P(-0.025 \leq Z \leq 1.25)$$

$$= P(-0.025 \leq Z \leq 0) + P(0 \leq Z \leq 1.25)$$

$$\text{from table, } P(75 \leq \bar{X} \leq 78) = 0.2324 + 0.2449$$

$$= 0.4773$$

Q15) A car hire firm has two cars, which it hires out day by day. The no. of demands for a car on each day distributed as a Poisson distribution with mean 1.5. calculate the proportions of days on which (i) neither car is used and (ii) the proportion of days on which some demand is made.

$$\lambda = 1.5$$

$$P(X=0) = \frac{\lambda^0 e^{-\lambda}}{0!}$$

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(i) Neither car is used

$$P(n=0) = \frac{1.5^0 e^{-1.5}}{0!} = \frac{1 \times e^{-1.5}}{1} = e^{-1.5} = 22.31$$

$$\therefore 22.31\%$$

(ii) some demand surplus

we need to find prob that the demand for car exceeds the no. of available cars.

$$P(n > 2) = 1 - P(n \leq 2)$$

$$P(n > 2) = 1 - (P(n=0) + P(n=1) + P(n=2))$$

$$P(n > 2) = 1 - \left(\frac{1.5^0 e^{-1.5}}{0!} + \frac{1.5^1 e^{-1.5}}{1!} + \frac{1.5^2 e^{-1.5}}{2!} \right)$$

$$= 1 - 0.889$$

$$= 0.112$$

$$= 11.2\%$$

(iii) Assume mean height of soldier to be 68.22 with variance of 10.8 inches. How many soldiers in a Regiment of 1000 would you expect to be over 6 feet tall, given that the area under the standard normal curve b/w $z=0$ and $z=0.35$ is 0.1368 and b/w $z=0$ and $z=1.5$ is 0.3749

$$z = \frac{n - \bar{x}}{s} = \frac{72 - 68.22}{\sqrt{10.8}} = 1.15$$

$$\text{Area } z=1.15 = 0.5 - 0.3749 = 0.125$$

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$$\sigma^2 = 10.8 \quad \sigma = \sqrt{10.8} = 3.286$$

$$n = 72 \quad \bar{x} = 68.22$$

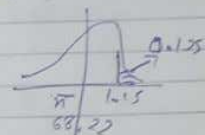
$$z = \frac{n - \bar{x}}{\sigma} = \frac{72 - 68.22}{3.286} = 1.15$$

$$\text{Area } 0 \text{ to } 1.15 = 0.3749$$

$$\text{Area under curve} = 0.5 - 0.3749$$

$$= 0.125$$

$$\text{for 1000 soldiers our sin fact} = 0.125 \times 1000 = 125$$



(i) If x is uniformly distributed in $[-2, 5]$ find

$$(i) P(x < 0)$$

$$(ii) P(x-1 \geq 1/2)$$

$$f(x) = \frac{1}{b-a} = \frac{1}{5-(-2)} = \frac{1}{7}$$

$$(i) P(x < 0) = \int_{-2}^0 f(x) dx$$

$$= \int_{-2}^0 \frac{1}{7} dx = \frac{1}{7} \times (0 - (-2)) = \frac{2}{7}$$

$$(ii) P(x-1 \geq 1/2) = P(x \leq 1/2) \cup P(x \geq 3/2)$$

$$\text{for } P(x \leq 1/2) = \int_{-2}^{1/2} \frac{1}{7} dx = \frac{1}{7} \times \left(\frac{1}{2} - (-2)\right) = \frac{5}{14}$$

$$P(x \geq 3/2) = \int_{3/2}^5 \frac{1}{7} dx = \frac{1}{7} \times (5 - 3/2) = \frac{7}{14} = \frac{1}{2}$$

$$P(x-1 \geq 1/2) = P(x \leq 1/2) + P(x \geq 3/2) = \frac{5}{14} + \frac{7}{14} = \frac{12}{14} = \frac{6}{7}$$

Date

- Q14) The time (in hrs) to repair a machine is exponentially distributed with parameter $\lambda = 1/3$. What is the prob that the repair time exceeds 3 hrs?

$$\lambda = 1/3 \quad P(T > 3) = ?$$

$$P(T > t) = e^{-\lambda t}$$

$$P(T > 3) = e^{-(1/3) \times 3} = e^{-1} = 0.3679$$

- Q15) The life length (in months) X of an electric component follows an exponentially distribution with $\lambda = 1/10$. What is the prob that the component survives at least 10 months given that already it had survived for more than 5 months?

$$\text{conditional probability} = P(X > 10 | X > 5)$$

$$P(A|B) = P(A \cap B) / P(B)$$

$$f(t) = 1 - e^{-\lambda t}$$

$$P(X \geq n) = 1 - P(X < n) = 1 - F(n) = e^{-\lambda n}$$

$$P(X \geq 10) = e^{-\lambda n} = e^{-(1/10) \times 10} = e^{-1}$$

$$P(X \geq 5) = e^{-(1/10) \times 5} = e^{-0.5}$$

$$P(X \geq 10 | X > 5) = P(X \geq 10) = e^{-1}$$

$$(i.e. X \geq 10 \text{ implies } X > 5)$$

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$$P(X \geq 10 | X > 5) = \frac{P(X \geq 10)}{P(X > 5)} = \frac{e^{-1}}{e^{-0.5}} = e^{-0.5} = 0.6065 = 60.65\%$$

- Q16) A random sample of size n is taken from a population whose mean is μ and variance is σ^2 . Using CLT, with what prob can we assert that the mean of the sample will not differ from $\mu = 60$ by more than 4?

$$\mu = 60$$

$$\sigma^2 = 400$$

$$\sigma = \sqrt{400} = 20$$

$$n = 100$$

prob that sample mean differs from population mean by at most 4

$$P(56 < \bar{x} < 64)$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{56 - 60}{20/\sqrt{100}} = -2$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\text{for } \bar{x} = 56 \Rightarrow \frac{56 - 60}{20/\sqrt{100}} = -2$$

$$\text{for } \bar{x} = 64 \Rightarrow \frac{64 - 60}{20/\sqrt{100}} = 2$$

$$P(Z < 2) = 0.9772$$

$$P(Z < -2) = 0.0228$$

from table

$$P(-2 < Z < 2) = P(Z < 2) - P(Z < -2)$$

$$= 0.9772 - 0.0228 = 0.9544$$

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Q17) In a normal distribution, 31% of items are under 45 and 8% are over 64. What is the mean and standard deviation of the distribution? $\phi(1.41) = 0.42$, $\phi(0.19) = 0.50$

$$z = \frac{x - \bar{x}}{\sigma}$$

$$-0.5 = \frac{45 - \bar{x}}{\sigma}$$

$$\Rightarrow -0.5\sigma = 45 - \bar{x}$$

$$\Rightarrow \bar{x} - 0.5\sigma = 45 \quad (1)$$

$$1.41 = \frac{64 - \bar{x}}{\sigma}$$

$$\bar{x} + 1.41\sigma = 64 \quad (2)$$

Solving eqn (1) and (2)

$$\bar{x} - 0.5\sigma = 45 \quad (1)$$

$$\bar{x} + 1.41\sigma = 64 \quad (2)$$

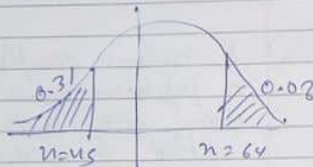
$$-1.91\sigma = -19 \Rightarrow \sigma = 10$$

$$\therefore \bar{x} = 0.5 \times 10 = 45$$

$$\Rightarrow \bar{x} = 50$$

$$\therefore \bar{x} = 50, \sigma = 10$$

Q18) A coin is tossed 12 times. Find the prob. both exactly and by fitting a normal distribution of getting (a) 4 heads (b) at most heads. Given $\phi(1.41) = 0.4251$, $\phi(0.866) = 0.51$



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Normal approximation works better for large values of n and p

$$p = 1/2 \quad n = 12$$

$$(i) P(4 \text{ heads}) = {}^{12}C_4 \left(\frac{1}{2}\right)^4 \times \left(\frac{1}{2}\right)^{12-4} = 0.117$$

(ii) at most 4 heads

$$P(0 \text{ heads}) + P(1 \text{ head}) + P(2 \text{ heads}) + P(3 \text{ heads}) + P(4 \text{ heads}) = P(\text{at most 4 heads})$$

$$= {}^{12}C_0 \left(\frac{1}{2}\right)^0 \times \left(\frac{1}{2}\right)^{12} + {}^{12}C_1 \left(\frac{1}{2}\right)^1 \times \left(\frac{1}{2}\right)^{11} + {}^{12}C_2 \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^{10} + {}^{12}C_3 \left(\frac{1}{2}\right)^3 \times \left(\frac{1}{2}\right)^9 + {}^{12}C_4 \left(\frac{1}{2}\right)^4 \times \left(\frac{1}{2}\right)^8$$

$$= 0.00244 + 0.00233 + 0.01709 + 0.05371 + 0.12085 = 0.19482$$

Q19) The no. of PCs sold daily at computer world is uniformly distributed with a min of 2000 PCs and a max of 5000 PCs

(i) Find the prob. that the daily sale will fall between 2500 and 3000 PCs.

(ii) what is the prob. that comp. world will sell at least 1000 PCs and exactly 2000 PCs?

$$(i) f(x) = \frac{1}{b-a} = \frac{1}{5000-2000} = \frac{1}{3000}$$

$$P(2500 \leq x \leq 3000) = \text{length of interval} \times \text{Prob. density} = 500 \times \frac{1}{3000} = \frac{1}{6} = 0.1667$$

$$\begin{aligned}
 \text{(ii)} \quad P(X \geq 4000) &= \text{length of interval} \times \text{prob density} \\
 &= (5000 - 4000) \times \frac{1}{3000} \\
 &= \frac{1}{3} = 0.333
 \end{aligned}$$

Prob of exactly 2500 $P(X = 2500) = 0$
 $\because$ this is continuous distribution,
 area under prob curve at a single point
 is 0 in a continuous distribution.

Q20) The length of a time a person speaks over
 phone follows exponential distribution with
 mean 6. What is the prob. that the
 person will talk for.

- (i) more than 8 min
- (ii) b/w 4 and 8 min

$$\text{(i)} \quad P(T > 8)$$

$$\lambda = 1/\mu = 1/6$$

$$P(T > 8) = e^{-\lambda \times 8} = e^{-1/6 \times 8} = e^{-4/3} = 0.2636$$

$$\text{(ii)} \quad P(4 < T < 8) = P(T < 8) - P(T < 4)$$

$$P(T < t) = 1 - e^{-\lambda t}$$

$$P(T < 8) = 1 - e^{-1/6 \times 8} = 1 - e^{-4/3} = 0.7364$$

$$P(T < 4) = 1 - e^{-1/6 \times 4} = 1 - e^{-2/3} = 0.4866$$

$$\begin{aligned}
 P(4 < T < 8) &= P(T < 8) - P(T < 4) = 0.7364 - 0.4866 \\
 &= 0.2498
 \end{aligned}$$